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CS373

HW3 Network Traffic Analysis.

1. Done
2. After executing the stat command on network R, I observed a high volume of requests to TCP port 139, which is associated with SMB. Additionally, there were significant requests to port 80, 110, and UDP port 53. Port 80 handles web traffic, TCP port 110 is used for email, and port 53 is for DNS. Given the types of traffic recorded in the network log, it appears that this network is likely an office environment where users are engaged in activities like reading emails and sharing files.
   In network O, I observed a substantial number of requests to ports 25, 80, and UDP port 500. Additionally, while TCP port 445 shows some traffic, it is relatively minimal. Port 25 is used for SMTP, and given that there are over 200,000 requests to this port, it is highly probable that this network is intended for email services.
3. OK
4. In network R, I observed that the predominant network prefix was 10.5.63.xxx. A brief search indicates that this address range is designated for use within private networks. This pattern of traffic suggests that the network is likely used for an office environment. In network O, the IP address 199.249.33.xxx was predominant in the logs, with 207.182.xx.xxx also being a significant presence. Since these are not private IP addresses, it is possible that they are involved in handling email traffic.
5. 10..5.63.xxx dominated the traffic on network R and 199.249.33.xxx dominated network O.
6. A
7. Network R didn't have any logs with any of these protocols in it. This seems indicative of a network that primarily handles local traffic. There were only a few requests using the OSPF protocol ID, suggesting that network O likely does not rely heavily on routers for handling requests. Additionally, most requests using protocol 89 came from a single IP address. There were also very few requests from the IPSEC protocol, and these were spread across various addresses. Furthermore, requests from the GRE protocol version were even scarcer. Overall, it seems that network O has minimal direct user interaction, similar to an email server.
8. I think it is helpful but only one piece of information that informs on the greater situation.
9. OK
10. R:

32.98.255.112

234.42.42.42

234.142.142.142

216.101.171.2 - Email Server

209.67.181.20

209.67.181.11

208.10.192.202

208.10.192.176

208.10.192.175

208.10.192.161

207.5.63.20

207.5.63.2 - DNS

207.46.143.254

207.44.165.251 - Email

206.253.217.13

206.170.168.217 - Printer

206.13.28.62 - Email

204.71.201.113

204.71.200.246

For network O my top twenty IP's didn't have any interesting ports listed for them.

11.

IP's don't seem to be related